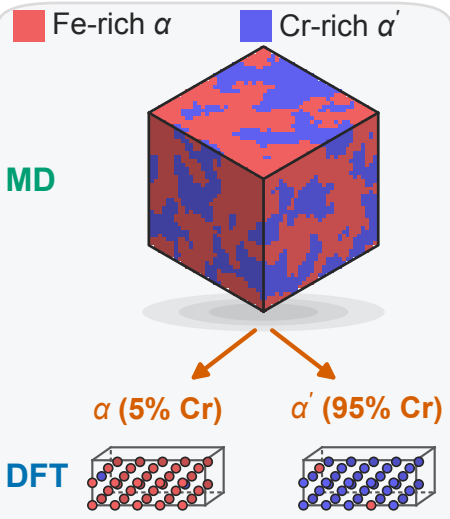
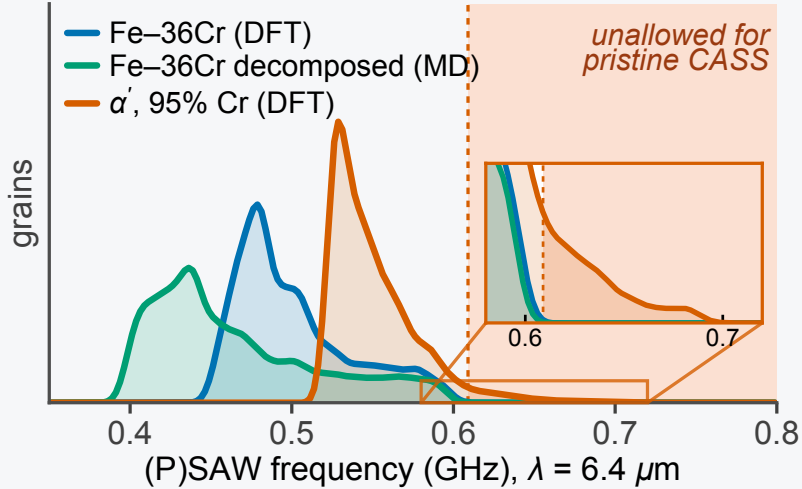
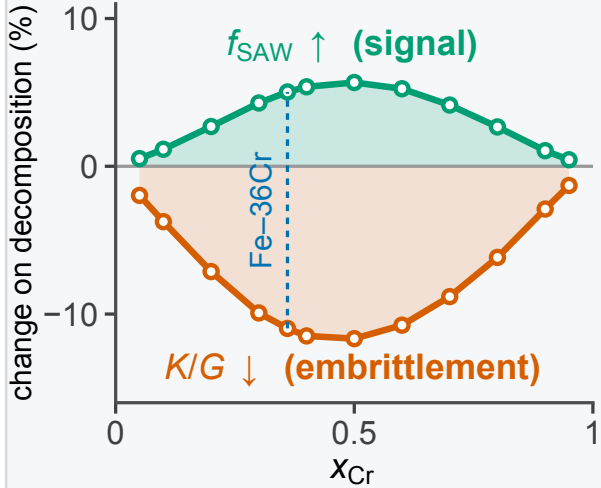



**Simulated spinodal decomposition of Fe–36Cr**




**DFT: stiff Cr-rich α′ places surface waves in the range unallowed for pristine ferrite**




**Shear stiffening raises $f_{SAW}$ and lowers $K/G$: a signature of embrittlement**